

INVESTIGATING APPROXIMATIONS AND PARAMETER ESTIMATION OF THE MULTIVARIATE GENERALIZED BURR-GAMMA

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Key words: Approximation; Estimation; Extreme Values; Kolmogorov-Smirnov; MGBG; QQ-plot.

Summary: The MGBG is considered for modelling multivariate data especially containing extreme values. This article is an extension and comparison to an earlier article by Beirlant et al. (2000). Special attention is given to the approximations of the expected value and covariance of the MGBG and to the estimation of parameters. The Kolmogorov-Smirnov goodness-of-fit measure is considered for estimating some of the parameters.

1. Introduction

The multivariate Generalized Burr-Gamma (MGBG) distribution (Beirlant et al., 2000) generalizes the Burr-Gamma distribution (Beirlant et al., 2002) to a multivariate distribution which is fairly flexible to fit to multivariate data containing extremes on all or some of the variables. In this paper an alternative approach to Beirlant et al. (2000) is taken to estimate the parameters of the MGBG. In Section 2 properties of the MGBG are given and in Section 3 asymptotic formulae are derived for $E(\underline{X})$ and $Var(\underline{X})$. These results are tested for a few simulated data sets in Section 4. In Section 5 we discuss

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our alternative approach to estimate the parameters and then the procedure is applied to a simulated data set in Section 6. Section 7 shows the difference in the estimated parameter values between the approach in Beirlant et al. (2000) and our approach.

2. The Multivariate Generalized Burr-Gamma distribution

This section gives a brief introduction to the Multivariate Generalized Burr-Gamma distribution and discusses some of the basic properties of the Multivariate Generalized Burr-Gamma.

The Multivariate Generalized Burr-Gamma distribution (MGBG) is an extension of the univariate generalized Burr-Gamma family of distributions and it allows the modelling of multivariate data that includes extreme values.

A random vector $\underline{X} = (X_1, \dots, X_p)'$ is MGBG($\underline{k}, \underline{\mu}, \Sigma, \underline{\xi}$) distributed if the joint density function is given by the following equation:

$$f(\underline{x}) = \frac{1}{|\Sigma|^{\frac{1}{2}}} \prod_{i=1}^p \frac{\sqrt{\psi' k_i}}{\Gamma(k_i) x_i} \exp(-\nu_{\xi_i}(\underline{x})) \nu_{\xi_i}(\underline{x})^{k_i-1} (1 + \xi_i \nu_i(\underline{x}))^{-1} \nu_i(\underline{x}),$$

$$\cdot 1 + \xi_i \nu_i(\underline{x}) > 0$$

$$= \prod_{i=1}^p \Gamma(k_i, \nu_{\xi_i}(\underline{x}))$$
(1)

where

$$V_{\xi_i} = \frac{1}{\xi_i} \log(1 + \xi_i \nu_i) \sim GAM(k_i, 1), \text{ independent for } i = 1, \dots, p$$

$$V_i = \exp \left\{ \psi(k_i) - \sqrt{\psi'(k_i)} \sum_{(i)}^{\frac{-1}{2}} (\underline{Y} - \underline{\mu}) \right\},$$

$$\begin{aligned}\underline{Y} &= (Y_1, \dots, Y_p)', \quad Y_i = -\log(X_i), \\ \psi(k_i) &= \frac{\partial}{\partial k_i} \log \Gamma(k_i) \quad \text{and} \quad \psi'(k_i) = \frac{\partial}{\partial k_i} \psi(k_i).\end{aligned}$$

$\psi(k_i)$ and $\psi'(k_i)$ represent the digamma and trigamma functions respectively. For dimension p , $\sum_{(i)}^{-1}$ is the i^{th} row of the symmetric square root matrix Σ^{-1} , where Σ is a symmetric positive definite $p \times p$ matrix. $\underline{k} = (k_1, \dots, k_p)$ denotes a positive vector of shape parameters, $\underline{\mu} = (\mu_1, \dots, \mu_p)$ denotes a vector of location parameters and $\underline{\xi} = (\xi_1, \dots, \xi_p)$ denotes a vector of extreme value indices. The parameter space is defined as

$$\Omega = \{k_i > 0, \quad -\infty < \mu_i < \infty, \quad \Sigma > 0, \quad -\infty < \xi_i < \infty\}, \quad i = 1, \dots, p$$

(Beirlant *et al.* 2000, p. 113).

Remark. If $\xi_i = 0$ for $i = 1, \dots, p$, then $\mu_i = E(Y_i)$ and $\Sigma = \text{Cov}(\underline{Y})$. The implication of the remark is that $\underline{\mu}$ and Σ can be estimated from the data by deleting the extreme observations exceeding certain thresholds t_i , $i = 1, \dots, p$. This implies that $\underline{\mu}$ and Σ are not the mean and covariance of $\underline{X}$. The first question that we address is: what is $E(\underline{X})$ and $\text{Cov}(\underline{X})$?

3. Approximations for $E(\underline{X})$, $\text{VAR}(\underline{X})$ and $\text{COV}(\underline{X})$

In this section approximated formulae are derived for $E(\underline{X})$, $\text{VAR}(\underline{X})$ and $\text{COV}(\underline{X})$, these approximations are based on Taylor expansions. For a more detailed explanation refer to the Appendix.

The approximated expected value of $\underline{X}$ is

$$E(X_i) \approx \exp(-\mu_{Y_i}) + \frac{1}{2} \sigma_{Y_i}^2 \exp(-\mu_{Y_i}) \quad (2)$$

where

$$\mu_{Y_i} = E(Y_i) = -\sum_{(i)}^{\frac{1}{2}} D_{\psi}^{-1} E \begin{bmatrix} \log V_1 \\ \cdot \\ \cdot \\ \cdot \\ \log V_p \end{bmatrix} + \sum_{(i)}^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \psi(k_1) \\ \cdot \\ \cdot \\ \cdot \\ \psi(k_p) \end{bmatrix} + \mu_i ,$$

$D_{\psi} = \text{diag}(\sqrt{\psi'(k_1)}, \sqrt{\psi'(k_2)}, \dots, \sqrt{\psi'(k_p)})$ and $\sigma_{Y_i}^2$ is the i^{th} diagonal element of

$$\text{Cov}(\underline{Y}, \underline{Y}') = \sum^{\frac{1}{2}} D_{\psi}^{-1} \text{Cov}(\log V, \log V') D_{\psi}^{-1} \sum^{\frac{1}{2}} .$$

This approximation is proven in Appendix A.1.

The approximated variance of $\underline{X}$ is

$$\text{Var}(X_i) \approx k_i \left[\exp \left(\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{pmatrix} \log \left(\frac{e^{\xi_1 V_{\xi_1}}}{\xi_1} \right) - \psi(k_1) \\ \cdot \\ \cdot \\ \cdot \\ \log \left(\frac{e^{\xi_p V_{\xi_p} - 1}}{\xi_p} \right) - \psi(k_p) \end{pmatrix} - \mu_i \right) \right. \\ \left. \left(\left[\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \left(\frac{\xi_j}{e^{\xi_j k_j} - 1} \right) e^{\xi_j k_j} \right] \right)^2 \right] . \quad (3)$$

This approximation is proven in Appendix A.2.

The approximated covariance between X_i and $X_{i'}$ is

$$\text{Cov}(X_i, X_{i'}) \approx a_i \text{Cov}(V_{\xi_{i'}}, V_{\xi_i}) (a_{i'})' \quad (4)$$

where

$$a_i = \exp \left[\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{pmatrix} \log \frac{e^{\xi_1 k_1} - 1}{\xi_1} - \psi(k_1) \\ \vdots \\ \log \left(\frac{e^{\xi_p k_p} - 1}{\xi_p} \right) - \psi(k_p) \end{pmatrix} - \mu_i \right] \\ \times \sum_i^{\frac{1}{2}} D_{\psi}^{-1} \frac{\xi_i e^{\xi_i k_i}}{e^{\xi_i k_i} - 1}$$

and $Cov(V_{\underline{\xi}}' V_{\underline{\xi}}')$ is a diagonal matrix with $Var(V_{\xi_i}) = k_i$ on the diagonal.

This approximation is proven in Appendix A.3.

From (2), (3) and (4) it is clear that $E(\underline{X})$, $Var(\underline{X})$ and $Cov(\underline{X})$ are complicated functions of the parameters. To estimate them, we propose the estimation of the parameters and substitute the estimates in these functions. In the next section we will investigate these estimations through simulations.

4. Investigating the approximations

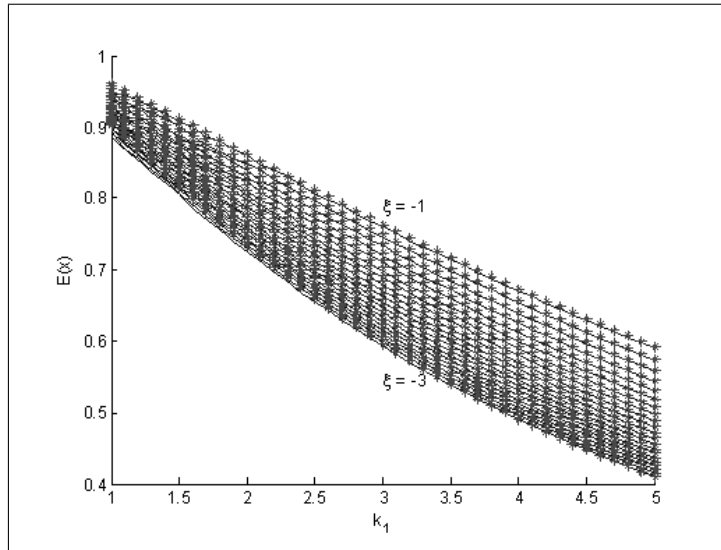
In this section we investigate the appropriateness of the approximations for $E(\underline{X})$ and $Var(\underline{X})$ in (2) and (3). A three-dimensional data set with specific parameter values was simulated from a MGBG distribution and it is investigated for which values of k_i and ξ_i , $i = 1, 2, 3$ the approximated mean and variance are close to the estimated mean and variance.

After simulating a three-dimensional data set of size $n = 500$ from a MGBG distribution the approximated mean and variance are compared to the estimated mean and variance of $\underline{X}$. First 500 $GAM(k_i, 1)$ values are simulated and then transformed to X , using the transformations from Section 2. This is illustrated in the following figures. In Figures 1 and 2 the estimated mean

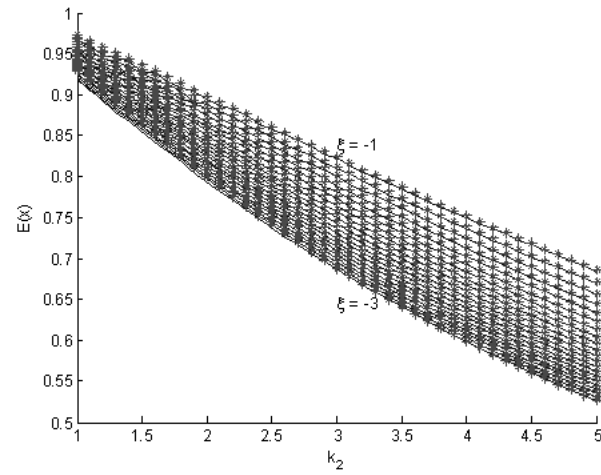
and variance are indicated by the solid line and the approximated mean and variance are indicated by '*' for each variable. For the simulations $\mu = [0 \ 0 \ 0]$ and $\Sigma = \begin{bmatrix} 0.014 & 0.004 & 0.007 \\ 0.004 & 0.007 & 0.004 \\ 0.007 & 0.004 & 0.013 \end{bmatrix}$ are assumed. Figures 1 and 2 show that for $1 \leq k_i \leq 5$ and $-3 \leq \xi_i \leq -1$, $i = 1, 2, 3$, the approximated mean and variance of $\underline{X}$ are rather close to the estimated mean and variance of $\underline{X}$ for each dimension in this simulated data set. For this simulated data set it can be shown that for $\xi_i > 1.5$ the estimated variance becomes rather volatile.

Figure 1. Estimated $E(\underline{X})$ (indicated by -) and the approximated $E(\underline{X})$ (indicated by *) for the three different dimensions plotted against k .

(a) Variable 1



(b) Variable 2



(c) Variable 3

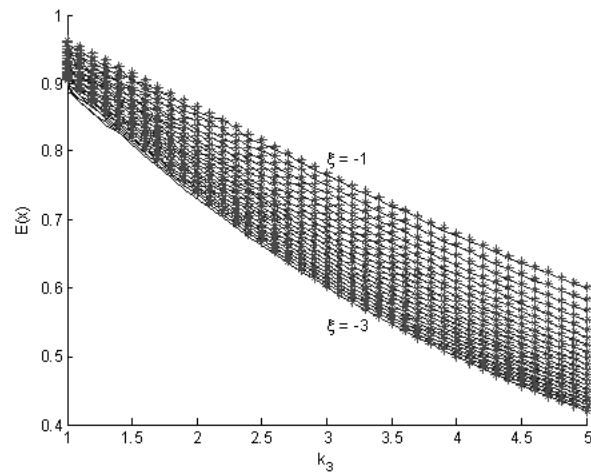
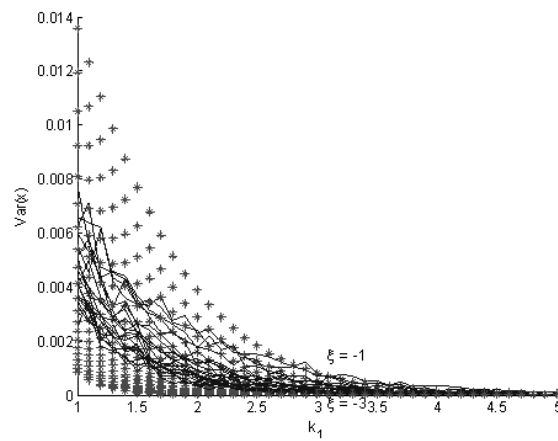
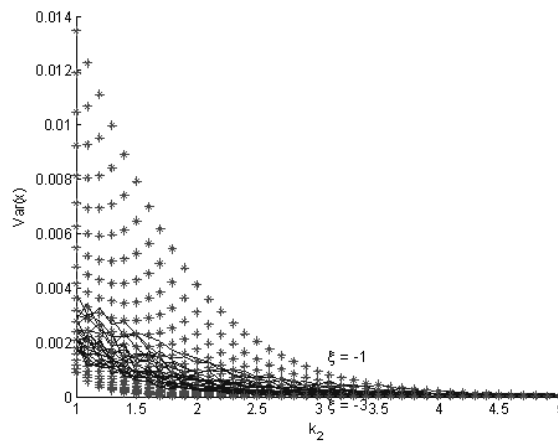


Figure 2. Estimated $Var(\underline{X})$ (indicated by -) vs. the approximated $Var(\underline{X})$ (indicated by *) for the three dimensions.

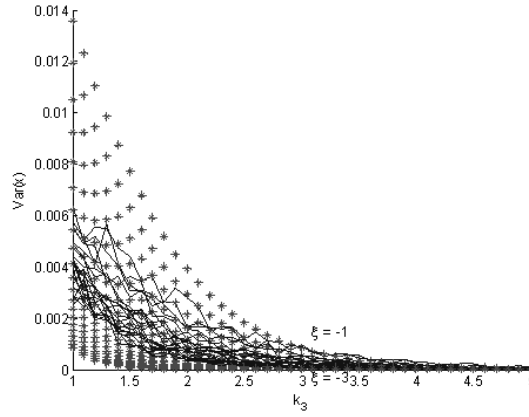
(a) Variable 1



(b) Variable 2



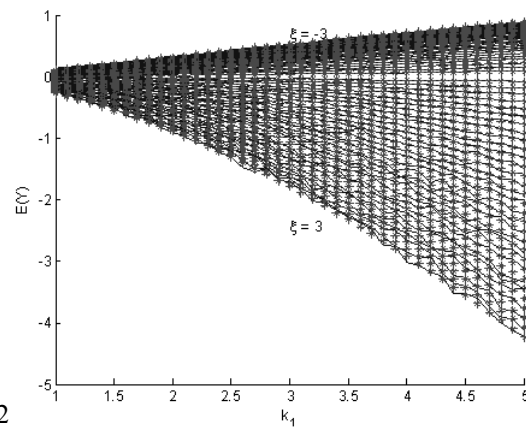
(c) Variable 3



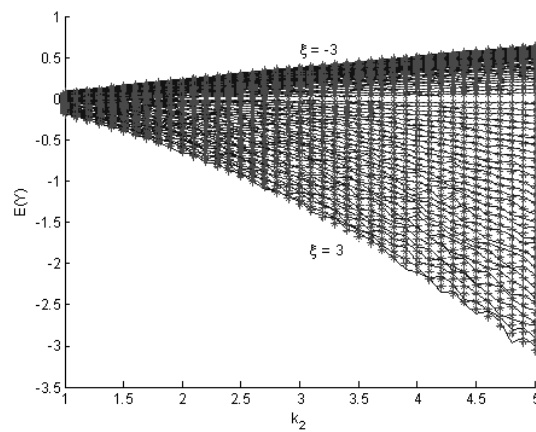
Similar simulations can be done where the estimated $E(\underline{Y})$ is compared to the approximated $E(\underline{Y})$ (given in the Appendix). The estimated mean and variance are indicated by the solid line and the approximated mean and variance are indicated by '*'. Figures 3 and 4 show the comparison for $1 \leq k_i \leq 5$ and $-3 \leq \xi_i \leq 3$, $i = 1, 2, 3$. From the figures it can be seen that the approximated mean and variance of $\underline{Y}$ are close to the estimated mean and variance of $\underline{Y}$ for each dimension for this simulated data set.

Figure 3. Estimated $E(\underline{Y})$ (indicated by -) vs. the approximated $E(\underline{Y})$ (indicated by *) for the three dimensions plotted against k .

(a) Variable 1



(b) Variable 2



(c) Variable 3

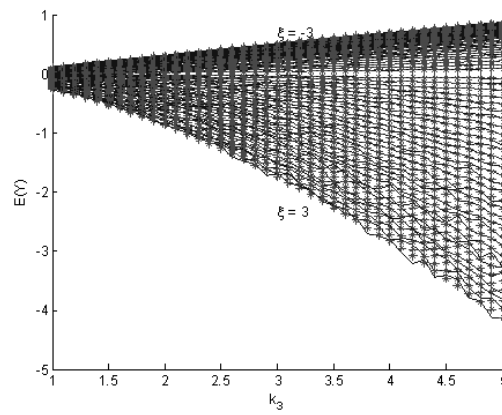
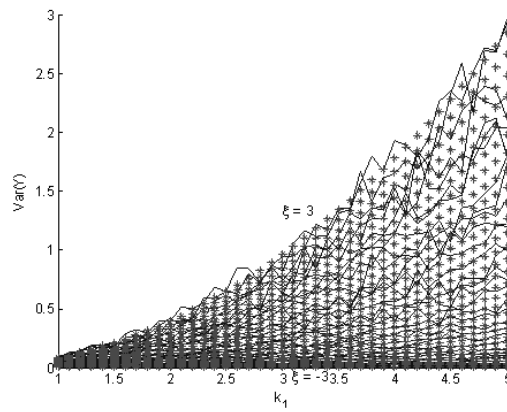
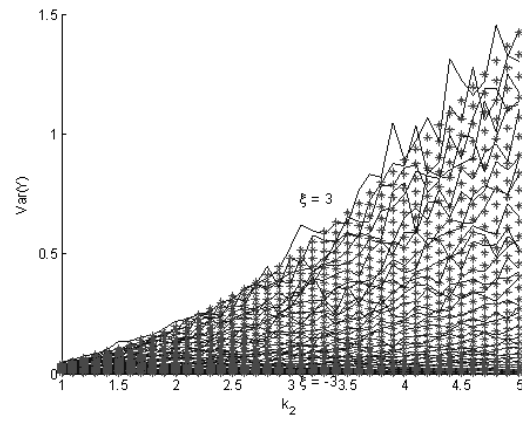


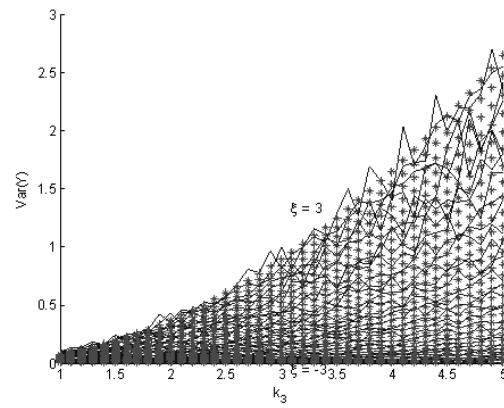
Figure 4. Estimated $Var(\underline{Y})$ (indicated by -) vs. the approximated $Var(\underline{Y})$ (indicated by *) for the three dimensions.
(a) Variable 1



(b) Variable 2



(c) Variable 3



5. Estimation of parameters $\underline{k}$ and $\underline{\xi}$ through the KS measure

This section discusses the estimation of the four MGBG parameters. Here we take a different approach for estimating $\underline{k}$ and $\underline{\xi}$, the Kolmogorov-Smirnov measure.

The approach for estimating the parameters $\underline{\mu}$ and Σ are the same as in the article of Beirlant *et al.* $\underline{\mu}$ and Σ are estimated by first “trimming” the data, thus ignoring the extreme values, and then using the method of moments to estimate $\underline{\mu}$ and Σ from the data below a threshold. The method of moments is given by the following equations

$$\widehat{\underline{\mu}} = \sum_{j=1}^{n_t} (y_{1j}, \dots, y_{pj})' / n_t \quad (5)$$

and

$$\widehat{\Sigma} = \left\{ \sum_{j=1}^{n_t} (y_{ij}, \dots, y_{pj})' (y_{ij}, \dots, y_{pj}) - \widehat{\underline{\mu}} \widehat{\underline{\mu}}' \right\} / n_t, \quad i = 1, \dots, p; \quad j = 1, \dots, n_t \quad (6)$$

where n_t denotes the number of observations below the threshold t . Thresholds are chosen for each dimension by obtaining the 75th upper quartiles. The 75th upper quartile is chosen here as the thresholds to be consistent with the article of Beirlant *et al.* (2000).

The Kolmogorov-Smirnov measure, $KS = \max |F_n - F|$ (Conover, 1980), is then used to estimate values for $\underline{k}$ and $\underline{\xi}$ where F_n denotes the empirical cdf and F the fitted cdf. Since it is known that $V \sim GAM(k_i, 1)$ the Kolmogorov-Smirnov measure calculates the maximum absolute difference between the empirical Gamma function and the cumulative Gamma function for different values of $k_i > 0$ and $-\infty < \xi_i < \infty$, $i = 1, \dots, p$. With the KS

measure one can see how well the model fits the data. The minimum value of the different maximum KS measure values will indicate the best fit.

6. Simulated data sets

This section illustrates the estimation of the parameters as discussed in Section 5. First a data set with certain parameters is simulated and the parameters are estimated. The goodness of these estimated parameters are tested by drawing a QQ-plot. The simulation is then repeated a couple of times and the goodness of the estimated parameters is now tested by calculating the mean squared error for every repeated simulation.

$n = 500$ values x_{ij} , $i = 1, 2, 3$; $j = 1, \dots, n$ were simulated from a MGBG with the following set of parameters,

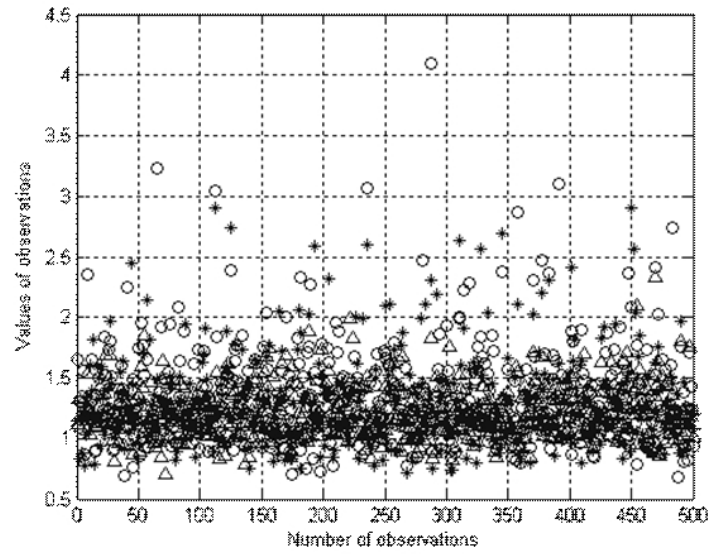
$$\mu = [0 \ 0 \ 0], \Sigma = \begin{bmatrix} 0.014 & 0.004 & 0.007 \\ 0.004 & 0.007 & 0.004 \\ 0.007 & 0.004 & 0.013 \end{bmatrix}, k = [1.7 \ 2.2 \ 2.1]$$

and

$$\xi = [1.2 \ 0.9 \ 1.0].$$

Figure 5 shows the simulated data values of X_{ij} , $i = 1, \dots, p$; $j = 1, \dots, n$. Let $Y_{ij} = -\log X_{ij}$, $i = 1, \dots, p$; $j = 1, \dots, n$.

Figure 5. Simulated value of $\underline{X}$, * indicates variable 1, $\triangle$ indicates variable 2 and o indicates variable 3.

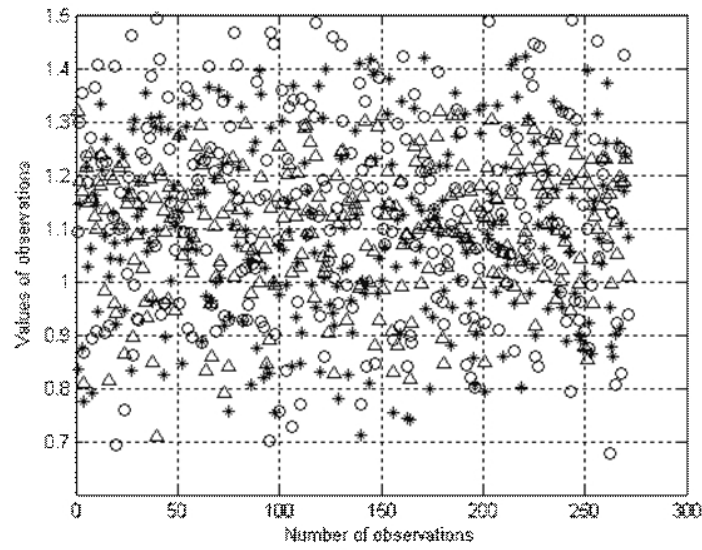


From Figure 5 it is evident that extreme values occur in the data set. The thresholds are now determined by obtaining the 75th upper quartile in every dimension.

Figure 6 shows the simulated $\underline{X}$ values below the thresholds. The mean and the variance of $\underline{Y}$ are now calculated for the data below the threshold as follows:

$$\hat{\underline{\mu}} = [-0.0731 \ -0.0897 \ -0.0997] \text{ and } \widehat{\underline{\Sigma}} = \begin{bmatrix} 0.0258 & 0.0043 & 0.0105 \\ 0.0043 & 0.0135 & 0.0054 \\ 0.0105 & 0.0054 & 0.0281 \end{bmatrix}.$$

Figure 6. Simulated value of $\underline{X}$ below the thresholds, * indicates variable 1, $\triangle$ indicates variable 2 and $\circ$ indicates variable 3.



For different values of $1 < k < 5$ and $1 < \xi < 1.9$ the KS measure is calculated. The estimates of $\underline{k}$ and $\underline{\xi}$ are the values of k and ξ that give the minimum KS measure value. The estimated parameter values are shown in Table 1 together with the minimum KS measure value. Table 1 also includes the estimated parameter values when the threshold is chosen as the 75th percentile.

Table 1. Estimated vs. true parameter values.

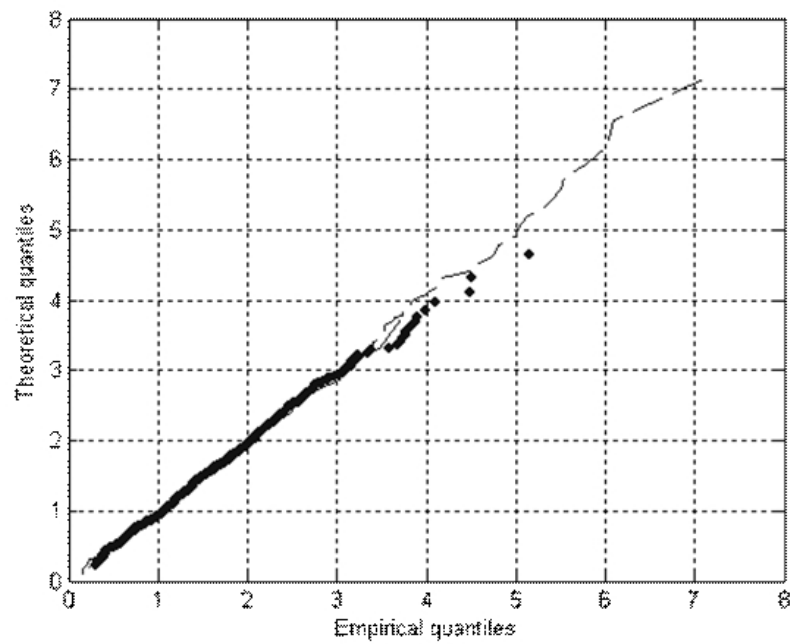
	True parameter value	Estimated parameter values for $t = 75^{\text{th}}$ quantiles
μ	[0 0 0]	[−0.0731 − 0.0897 − 0.0997]
Σ	$\begin{bmatrix} 0.014 & 0.004 & 0.007 \\ 0.004 & 0.007 & 0.004 \\ 0.007 & 0.004 & 0.013 \end{bmatrix}$	$\begin{bmatrix} 0.0258 & 0.0043 & 0.0105 \\ 0.0043 & 0.0135 & 0.0054 \\ 0.0105 & 0.0054 & 0.0281 \end{bmatrix}$
$\mathbf{k}$	[1.7 2.2 2.1]	[2.4 2.3 2.6]
ξ	[1.2 0.9 1.0]	[1.7 1 1.2]

The minimum KS values for each dimension are [0.0179 0.0236 0.0196], these values are all significant at the 5% level.

From Table 1 it is clear that the estimated parameter values are relatively close to the true parameter values.

Figure 7 shows the QQ-plots for each dimension for the estimated parameters in Table 1. A QQ-plot gives an indication of the goodness of fit of the MGBG to the simulated data. If the QQ-plot follows more or less the 45° straight line it indicates a good fit, as is the case in Figure 7.

Figure 7. QQ-plots for each dimension, dimension 1 is indicated by the solid line, dimension 2 by the dashed line and dimension 3 by the dots.



The above simulation is now repeated a 100 times and the mean squared error of the simulated parameter values and the estimated parameter values are calculated for each dimension. The MSE is given in the following table. The MSE between the eigenvalues of Σ and $\widehat{\Sigma}$ is calculated for each dimension.

Table 2. MSE values after 100 simulations.

	MSE, dimension 1	MSE, dimension 2	MSE, dimension 3
μ	0.0079	0.0075	0.0129
Σ	0.00004	0.0001	0.0003
$\mathbf{k}$	2.4198	2.0394	2.3458
ξ	0.1456	0.4774	0.3261

The table above shows that the MSE values are close to zero for all the parameter values except for k_i , $i = 1, \dots, p$.

Next we investigate the MSE of k_i , $i = 1, \dots, p$. Twenty data sets of size 500 are simulated for different values of $k_i = 1, 2$ or 3 , $i = 1, \dots, p$; while the rest of the parameters are kept constant, as given in Table 1. The MSE is calculated for every combination of k_i , $i = 1, \dots, p$. Only a limited combination of k_i values is taken here and the repetitions of simulated samples are reduced to 20 to reduce the execution time of the computer programme. The following table shows the MSE values for each dimension.

Table 3. MSE values of the different combinations of $k_i = 1, 2$ or 3 , $i = 1, \dots, p$.

	Dimension 1	Dimension 2	Dimension 3
$\underline{k} = [1 \ 1 \ 1]$	0.5615	0.421	1.702
$\underline{k} = [1 \ 1 \ 2]$	1.804	0.1145	1.7785
$\underline{k} = [1 \ 1 \ 3]$	0.527	0.822	0.3765
$\underline{k} = [1 \ 2 \ 1]$	2.01	1.862	0.9775
$\underline{k} = [1 \ 2 \ 2]$	1.156	1.4685	1.9885
$\underline{k} = [1 \ 2 \ 3]$	1.4835	2.1705	1.299
$\underline{k} = [1 \ 3 \ 1]$	2.128	0.868	0.5355
$\underline{k} = [1 \ 3 \ 2]$	0.184	1.083	1.2125
$\underline{k} = [1 \ 3 \ 3]$	2.721	1.089	1.046
$\underline{k} = [2 \ 1 \ 1]$	0.4895	1.0135	1.7565
$\underline{k} = [2 \ 1 \ 2]$	0.9155	0.616	2.365
$\underline{k} = [2 \ 1 \ 3]$	2.657	1.298	0.8625
$\underline{k} = [2 \ 2 \ 1]$	1.6345	2.211	1.3895
$\underline{k} = [2 \ 2 \ 2]$	0.832	1.151	2.172
$\underline{k} = [2 \ 2 \ 3]$	0.8915	2.3635	1.061
$\underline{k} = [2 \ 3 \ 1]$	1.1735	0.902	0.573
$\underline{k} = [2 \ 3 \ 2]$	1.282	1.5615	3.1615
$\underline{k} = [2 \ 3 \ 3]$	1.882	0.9175	1.021
$\underline{k} = [3 \ 1 \ 1]$	1.299	1.591	1.442
$\underline{k} = [3 \ 1 \ 2]$	0.6845	0.539	2.4115
$\underline{k} = [3 \ 1 \ 3]$	0.646	0.4395	0.9465
$\underline{k} = [3 \ 2 \ 1]$	1.473	1.1695	0.117
$\underline{k} = [3 \ 2 \ 2]$	0.6035	2.425	2.926
$\underline{k} = [3 \ 2 \ 3]$	0.9535	0.4985	1.2525
$\underline{k} = [3 \ 3 \ 1]$	0.727	0.9765	1.009
$\underline{k} = [3 \ 3 \ 2]$	0.8375	1.4675	2.2605
$\underline{k} = [3 \ 3 \ 3]$	0.9375	2.1825	1.538

From the above MSE values, which are relatively small, with variances also calculated as fairly small, we concluded that the estimates of k_i , $i = 1, \dots, p$ are acceptable.

7. Applications to real data

In this section the MGBG is fitted to a real data set and the parameters are estimated by using the methods in Section 5. The aim is now to compare our estimates with those of Beirlant *et al.* (2000), this is done by means of a QQ-plot.

The maximum monthly wind speed from March 1993 to December 1998, recorded in three stations in the Cape Town area, namely Cape Town harbour (HB), Cape Town airport (AP) and Robben Island (RI), are modelled with the MGBG. The same data set was considered previously by Beirlant *et al.* (2000) and the results are compared with our results. The data set is given in Appendix A4. Let $\underline{X} = (X_{HB}, X_{AP}, X_{RI})$ denote the wind speed in knots. The same thresholds were used as in the article by Beirlant *et al.* (2000), $t = (60, 55, 45)$. When considering only the data below the threshold $\underline{\mu}$, and $\underline{\Sigma}$ were estimated as

$$\hat{\underline{\mu}} = \begin{bmatrix} -3.8770 \\ -3.8873 \\ -3.5899 \end{bmatrix} \text{ and } \hat{\underline{\Sigma}} = \begin{bmatrix} 0.0135 & 0.0038 & 0.0068 \\ 0.0038 & 0.0070 & 0.0039 \\ 0.0068 & 0.0039 & 0.0124 \end{bmatrix}$$

which is the same as the results in Beirlant *et al.* (2000).

$\underline{\xi}$ and $\underline{k}$ were estimated simultaneously by using the Kolmogorov-Smirnov approach for different values of $0.3 < k_i < 3$ and $0.2 < \xi_i < 7$, $i = 1, 2, 3$, the range for k_i and ξ_i is chosen according to the article of Beirlant *et al.* (2000). The following estimates were obtained

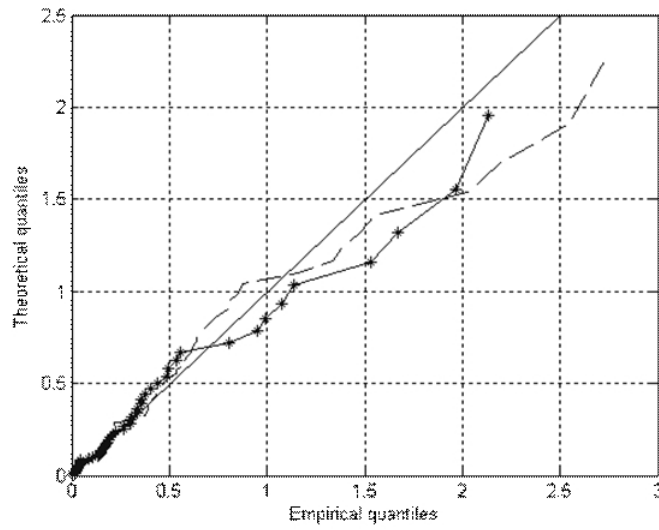
$$\hat{\underline{\xi}} = [6.2 \ 0.5 \ 1] \text{ and } \hat{\underline{k}} = [0.4 \ 0.8 \ 0.5].$$

The estimates of $\underline{\xi}$ and $\underline{k}$ differ considerably from the estimates in Beirlant *et al.* (2000) where $\hat{\underline{\xi}} = [2.8867 \ 1.3645 \ 0.3997]$ and $\hat{\underline{k}} = [0.8 \ 1.2 \ 1.3]$. In the article by Beirlant *et al.* (2000) $\underline{\mu}$ and $\underline{\Sigma}$ are also estimated through the method of moments, k_i , $i = 1, \dots, p$ is obtained by fitting a $GAM(k_i, 1)$ to

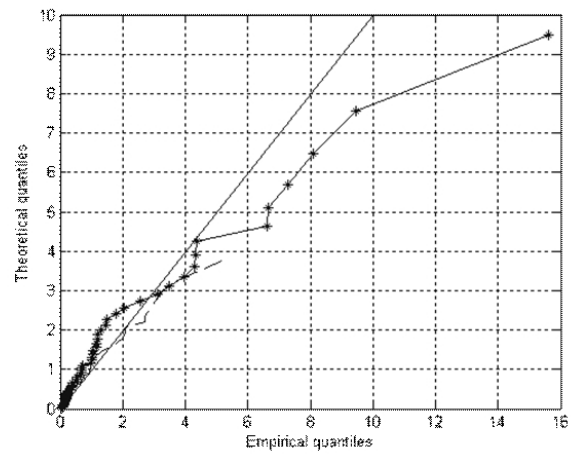
ν_{ij} , $i = 1, \dots, p$ and $j = 1, \dots, n_u$ where n_u is the number of values below the threshold by using QQ- or probability plots. ξ_i is then estimated through a Bayesian approach where all the data, including those above the threshold, are included. When estimating ξ_i , k_i is assumed to be fixed. QQ-plots with our parameters and the parameters of Beirlant *et al.* (2000) are shown in Figure 8 for the different dimensions.

Figure 8. QQ-plots for each dimension, (*-) represents the QQ-plots with our parameters and the (- -) represents the QQ-plots with the parameters of Beirlant *et al.* (2000) and the solid line represents the 45° line.

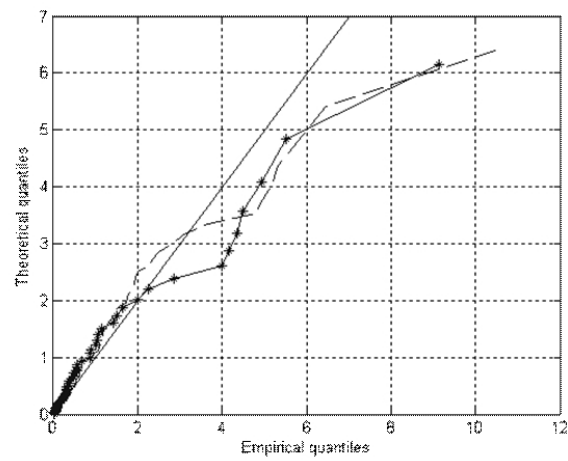
(a) Dimension 1



(b) Dimension 2



(c) Dimension 3



The line the closest to the 45° line will indicate the best fit. Although very similar, our model seems to be closer to the 45° line in figures (a) and (c). A further test for goodness of fit is the correlation coefficient r_Q between the empirical quantiles and the theoretical quantiles. The closer r_Q is to 1 the better the fit (Beirlant *et al.* 2004, p.9). The r_Q values are calculated as follows:

Table 4. Calculated r_Q values.

	Our model	Model of Beirlant <i>et al.</i> (2000)
Dimension 1	0.9895	0.9811
Dimension 2	0.9746	0.9856
Dimension 3	0.9797	0.9614

The correlation coefficients are high in both cases, although it is slightly higher in our model for dimensions 1 and 2.

8. Conclusion

The approach considered here for estimating the parameters of the MGBG seems to be appropriate and easier than previously reported. It seems to be a slight improvement on the estimation approach considered in Beirlant *et al.* (2000). Further research can be done by perhaps considering a thoroughly Bayesian approach for estimating all four parameters simultaneously.

Appendix

A.1:

The expected value of $X_i, i = 1, \dots, p$, where

$V_{\xi_i} = \frac{1}{\xi_i} \log(1 + \xi_i V_i) \sim \text{GAM}(k_i, 1)$ is given by the approximation

$$E(X_i) \approx \exp(-\mu_{Y_i}) + \frac{1}{2} \sigma_{Y_i}^2 \exp(-\mu_{Y_i}). \quad (\text{a.1})$$

Proof

From (a.1) $V_i = \frac{e^{\xi_i V_{\xi_i}}}{\xi_i}$, therefore $E\left[\log\left(\frac{e^{\xi_i V_{\xi_i}} - 1}{\xi_i}\right)\right] = E[\log(V_i)]$.

To obtain $E[\log(V_i)]$ the delta method given by Rice (1994, p. 149) is used. This method involves the expansion of the Taylor series to the second order to improve the approximation. Therefore $E(\log(V_i)) \approx g(\mu_{V_{\xi_i}}) + \frac{1}{2} \sigma_{V_{\xi_i}}^2 g''(\mu_{V_{\xi_i}})$,

where

$$\mu_{V_{\xi_i}} = k_i,$$

$$\sigma_{V_{\xi_i}}^2 = k_i,$$

$$g(\mu_{V_{\xi_i}}) = \log\left(\frac{e^{\xi_i \mu_{V_{\xi_i}}} - 1}{\xi_i}\right),$$

$$g'(\mu_{V_{\xi_i}}) = \frac{\xi_i e^{\xi_i \mu_{V_{\xi_i}}}}{e^{\xi_i \mu_{V_{\xi_i}}} - 1}$$

and

$$g''(\mu_{V_{\xi_i}}) = \frac{-\xi_i^2 e^{\xi_i \mu_{V_{\xi_i}}}}{(e^{\xi_i \mu_{V_{\xi_i}}} - 1)^2}.$$

Therefore

$$E(\log(V_i)) \approx \log\left(\frac{e^{\xi_i k_i} - 1}{\xi_i}\right) - \frac{k_i \xi_i^2 e^{\xi_i k_i}}{2(e^{\xi_i k_i} - 1)^2}. \quad (\text{a.2})$$

The delta method also indicates that $Var(\log V_i) \approx \sigma_{V_{\xi_i}}^2 \left(g'(\mu_{V_{\xi_i}})\right)^2$, therefore

$$Var(\log V_i) \approx k_i \left(\frac{\xi_i e^{\xi_i k_i}}{e^{\xi_i k_i} - 1}\right)^2. \quad (\text{a.3})$$

From (1) $Y_i = -\log(X_i)$, therefore $X_i = h(Y_i) = \exp(-Y_i)$, where

$$\begin{aligned} Y_i &= -\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{pmatrix} \log\left(\frac{e^{\xi_i V_{\xi_1}} - 1}{\xi_1}\right) - \psi(k_1) \\ \cdot \\ \cdot \\ \log\left(\frac{e^{\xi_p V_{\xi_p}} - 1}{\xi_p}\right) - \psi(k_p) \end{pmatrix} + \mu_i \\ &= -\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{pmatrix} \log(V_1) - \psi(k_1) \\ \cdot \\ \cdot \\ \log(V_p) - \psi(k_p) \end{pmatrix} + \mu_i \end{aligned} \quad (\text{a.4})$$

and

$$D_{\psi} = \text{diag}\left(\sqrt{\psi'(k_1)}, \sqrt{\psi'(k_2)}, \dots, \sqrt{\psi'(k_p)}\right). \quad (\text{a.5})$$

Again the delta method is used to obtain an approximation for $E(X_i)$ (Rice 1994, p. 149). Thus, when following the delta method, the following is obtained:

$$\begin{aligned} E(X_i) &\approx h(\mu_{Y_i}) + \frac{1}{2}\sigma_{Y_i}^2 h''(\mu_{Y_i}), \\ h'(Y_i) &= -\exp(Y_i) \text{ and } h''(Y_i) = \exp(-Y_i). \end{aligned} \quad (\text{a.6})$$

Therefore

$$E(X_i) \approx \exp(-\mu_{Y_i}) + \frac{1}{2}\sigma_{Y_i}^2 \exp(-\mu_{Y_i}) \quad (\text{a.7})$$

and

$$\mu_{Y_i} = E(Y_i) = -\sum_{(i)}^{\frac{1}{2}} D_{(\psi)}^{-1} E \begin{bmatrix} \log V_1 \\ \cdot \\ \cdot \\ \cdot \\ \log V_p \end{bmatrix} + \sum_{(i)}^{\frac{1}{2}} D_{(\psi)}^{-1} \begin{bmatrix} \psi(k_1) \\ \cdot \\ \cdot \\ \cdot \\ \psi(k_p) \end{bmatrix} + \mu_i \quad (\text{a.8})$$

where

$$E(\log(V_i)) \approx \log\left(\frac{e^{\xi_i k_i} - 1}{\xi_i}\right) - \frac{k_i \xi_i^2 e^{\xi_i k_i}}{2(e^{\xi_i k_i} - 1)^2}. \quad (\text{a.9})$$

$\sigma_{Y_i}^2$ is the i^{th} diagonal element of

$$\text{Cov}(\underline{Y}, \underline{Y}') = \sum^{\frac{1}{2}} D_{\psi}^{-1} \text{Cov}(\log V, \log V') D_{\psi}^{-1} \sum^{\frac{1}{2}}. \quad (\text{a.10})$$

Because V_{ξ_i} and $V_{\xi_{i'}}$ are independent, V_i and $V_{i'}$, $i \neq i'$ are also independent and therefore $\text{Cov}(\log V, \log V')$ is a diagonal matrix with $\text{Var}(\log V_i)$ on the

diagonal, where $\text{Var}(\log V_i) \approx k_i \left(\frac{\xi_i e^{\xi_i k_i}}{e^{\xi_i k_i} - 1} \right)^2$.

Therefore $E(X_i)$ is equal to

$$\begin{aligned}
& \exp \left[\left(\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \log \left(\frac{e^{\xi_1 k_1} - 1}{\xi_1} \right) - \frac{k_1 \xi_1^2 e^{\xi_1 k_1}}{2 (e^{\xi_1 k_1} - 1)^2} \\ \cdot \\ \cdot \\ \log \left(\frac{e^{\xi_p k_p} - 1}{\xi_p} \right) - \frac{k_p \xi_p^2 e^{\xi_p k_p}}{2 (e^{\xi_p k_p} - 1)^2} \end{bmatrix} \right) \right. \\
& \quad \left. - \sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \psi_1 \\ \cdot \\ \cdot \\ \cdot \\ \psi_p \end{bmatrix} - \mu_i \right] \\
& \quad + \frac{1}{2} \sigma_{Y_i}^2 \exp \left[\left(\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \log \left(\frac{e^{\xi_1 k_1} - 1}{\xi_1} \right) - \frac{k_1 \xi_1^2 e^{\xi_1 k_1}}{2 (e^{\xi_1 k_1} - 1)^2} \\ \cdot \\ \cdot \\ \cdot \\ \log \left(\frac{e^{\xi_p k_p} - 1}{\xi_p} \right) - \frac{k_p \xi_p^2 e^{\xi_p k_p}}{2 (e^{\xi_p k_p} - 1)^2} \end{bmatrix} \right) \right. \\
& \quad \left. - \sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \psi_1 \\ \cdot \\ \cdot \\ \cdot \\ \psi_p \end{bmatrix} - \mu_i \right] \tag{a.11}
\end{aligned}$$

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A.2:

The variance of X_i , $i = 1, \dots, p$, where

$V_{\xi_i} = \frac{1}{\xi_i} \log(1 + \xi_i V_i) \sim \text{GAM}(k_i, 1)$ is given by the approximation

$$\text{Var}(X_i) = k_i \left[\exp \left(\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{pmatrix} \log \left(\frac{e^{\xi_1 k_1} - 1}{\xi_1} \right) - \psi(k_1) \\ \cdot \\ \cdot \\ \log \left(\frac{e^{\xi_p k_p} - 1}{\xi_p} \right) - \psi(k_p) \end{pmatrix} \right) - \mu_i \right]^2$$

$$\left(\left[\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \left(\frac{\xi_i}{e^{\xi_i k_i} - 1} \right) e^{\xi_i k_i} \right] \right)^2 \quad (\text{a.12})$$

Proof

From A.1 Y_i is defined as $Y_i = - \sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{pmatrix} \log V_1 - \psi(k_1) \\ \cdot \\ \cdot \\ \log(V_p) - \psi(k_p) \end{pmatrix} + \mu_i$.

In A.1 it was shown that $X_i = \exp(-Y_i)$, $i = 1, \dots, p$, therefore

$$\begin{aligned}
 X_i &= \exp \left(\sum_i^{\frac{1}{2}} D_\psi^{-1} \begin{pmatrix} \log \left(\frac{e^{\xi_1 V_{\xi_1}} - 1}{\xi_1} \right) - \psi(k_1) \\ \vdots \\ \log \left(\frac{e^{\xi_p V_{\xi_p}} - 1}{\xi_p} \right) - \psi(k_p) \end{pmatrix} - \mu_i \right) \\
 &= g(V_{\xi_1}, \dots, V_{\xi_p})
 \end{aligned} \tag{a.13}$$

When applying the delta method

$$X_i \approx g(\mu_{V_{\xi_i}}) + (V_{\xi_i} - \mu_{V_{\xi_i}}) \frac{\partial g(\mu_{V_{\xi_i}})}{\partial V_{\xi_i}}. \tag{a.14}$$

Now it follows that

$$Var(X_i) \approx Var(V_{\xi_i}) \left(\frac{\partial g(\mu_{V_{\xi_i}})}{\partial V_{\xi_i}} \right)^2.$$

From A.1 we know that $\mu_{V_{\xi_i}} = k_i$ and $\sigma_{V_{\xi_i}}^2 = k_i$.

Therefore $Var(X_i)$ can be approximated by (a.12).

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A.3

The covariance of X_i , $i = 1, \dots, p$ where

$V_{\xi_i} = \frac{1}{\xi_i} \log(1 + \xi_i V_i) \sim \text{GAM}(k_i, 1)$ is given by the approximation

$$Cov(X_i, X_{i'}) = a_i Cov(V_{\xi_{i'}}, V_{\xi_i}') a_{i'}'. \tag{a.15}$$

Proof

From (a.14) $X_i \approx g(\mu_{V_{\xi_i}}) + (V_{\xi_i} - \mu_{V_{\xi_i}}) \frac{\partial g(\mu_{V_{\xi_i}})}{\partial V_{\xi_i}}$ and therefore

$$\begin{aligned} Cov(X_i, X_{i'}) &= E \left\{ \left[g(\mu_{\xi_i}) + (V_{\xi_i} - \mu_{V_{\xi_i}}) \frac{dg(\mu_{V_{\xi_i}})}{gV_{\xi_i}} - g(\mu_{V_{\xi_i}}) \right] \right. \\ &\quad \times \left. \left[g(\mu_{\xi_{i'}}) + (V_{\xi_{i'}} - \mu_{V_{\xi_{i'}}}) \frac{dg(\mu_{V_{\xi_{i'}}})}{gV_{\xi_{i'}}} - g(\mu_{V_{\xi_{i'}}}) \right] \right\}, \\ &\quad \text{for } i = 1, \dots, p \text{ and } i' = 1, \dots, p, \quad i \neq i' \end{aligned} \quad (\text{a.16})$$

$$\text{where } g(\mu_{\xi_i}) = \exp \left[\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \log \left(\frac{e^{\xi_1 V_{\xi_1} - 1}}{\xi_1} \right) - \psi(k_1) \\ \vdots \\ \log \left(\frac{e^{\xi_p V_{\xi_p} - 1}}{\xi_p} \right) - \psi(k_p) \end{bmatrix} - \mu_i \right] \text{ and}$$

$$\mu_{V_{\xi_i}} = k_i.$$

$$\text{For } a_i = \exp \left[\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \begin{bmatrix} \log \left(\frac{e^{\xi_1 k_1} - 1}{\xi_1} \right) - \psi(k_1) \\ \vdots \\ \log \left(\frac{e^{\xi_p k_p} - 1}{\xi_p} \right) - \psi(k_p) \end{bmatrix} - \mu_i \right]$$

$$\sum_i^{\frac{1}{2}} D_{\psi}^{-1} \frac{\xi_i e^{\xi_i k_i}}{e^{\xi_i k_i} - 1} \text{ the}$$

$$Cov(X_i, X_{i'}) = a_i Cov(V_{\xi_{i'}} V_{\xi_i}') a_{i'}'. \quad (\text{a.17})$$

Because V_{ξ_i} and $V_{\xi_{i'}}$ are independent, $Cov(V_{\xi_i} V_{\xi_i}')$ is a diagonal matrix with $Var(V_{\xi_i}) = k_i$ in the diagonal.

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A.4

Maximum monthly wind speed data measured at Cape Town Harbour, Cape Town Airport and Robben Island, from March 1993 to December 1998.

Harbour	Airport	Robben Island
121	45	35
49	56	37
39	41	27
41	45	35
56	60	39
56	47	37
39	49	41
49	43	37
53	47	39
53	49	43
51	47	39
51	47	39
47	43	39
60	45	37
45	45	37
89	70	53
49	54	39
43	45	43
41	45	33
43	39	31
43	45	33
47	47	37
56	51	37
49	43	35
62	41	35
51	47	64
136	51	33
70	56	45
89	51	33
58	45	41

Harbour	Airport	Robben Island
41	45	31
54	54	45
45	47	37
54	53	41
54	49	39
49	51	31
53	47	35
49	47	31
37	45	29
51	64	41
47	49	33
97	49	41
47	53	37
56	47	41
53	51	41
51	53	37
53	49	37
58	51	39
54	45	37
47	39	41
49	47	31
58	70	45
37	43	33
72	51	41
53	45	45
45	45	37
43	49	37
56	54	41
54	51	39
54	49	39
49	45	33
47	37	29
51	47	39
43	41	29
43	47	39
45	47	35

Harbour	Airport	Robben Island
51	47	33
49	41	37
53	43	39
54	43	39

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